A068 v0.4.4 — PKLSA Intrinsic Phase Clock Material Tubes, Full-field RPO and Floquet Falsifier

SST-Workbench | Framework v1.0.10

19 October 2026 – development / physical inference unresolved

Abstract

This version *actually evolves* a fully three-dimensional unforced incompressible Euler velocity field initialized from a SHA-256-identified native KnotPlot trefoil, using the unchanged A064 Euler solver, finite-core regularization, Fourier truncation and RK4. Reconstructed 3D vorticity yields an enstrophy-weighted shape coordinate and a separate enstrophy/energy functional; neither has an imposed frequency. A new Lagrangian torus is advected through the same RK4 stages as the actual Euler field, with local vorticity-tangency and material-volume diagnostics. Complete 3D velocity fields are searched for returns under three periodic translations and the 24 proper cubic rotations. The exact A064 tangent operator is separately finite-difference verified; Floquet Ritz prediction is strictly conditional on RPO and material-tube screening. Space/time/core refinement and framework provenance are retained. A separately preregistered 2015 notebook A/B geometric seed atlas is additionally audited and used for controlled short Euler pilot evolutions, without promoting source shape differences to physical-provider independence. This is a **numerical model** initialized by PKLSA/KnotPlot geometry; it is *not* a confirmed source-native material vortex or an independent physical PKLSA clock. Previously inspected geometry and past diagnostics remain **unblinded development evidence**; official BLIND packaging does not retroactively restore epistemic blindness.

1 Identification and certification boundaries

Field	Value
Catalog	A068
Version	v0.4.4
Framework	SST Falsifier Framework v1.0.10 candidate
Profile	`multilibrary_gpu`
Current authority	`EULER_MODEL_SIMULATION_ONLY`
Physical clock	`UNRESOLVED`

2 Research question and objective

Can an autonomous, unforced 3D Euler vortex initialized from native trefoil geometry generate a reproducible intrinsic phase without inserting a clock frequency? What additional independent tests are needed to claim a *material, physical PKLSA* clock? The present version is a numerical prerequisite, not a physical verification.

3 Competing hypotheses and decision semantics

H_0: a numerical transient, slow vortex deformation, rigid-body phase, inadequate sampling, nonconvergent core/mesh effects, or operator-fitting artefact mimics a clock. H_1: a genuine source-native material vortex has a stable relative periodic orbit, coherent intrinsic oscillatory phase and independent withheld operator prediction. An unqualified source produces `UNRESOLVED`, not physical `FAIL`; a failed numerical discretization is not an SST-theory falsification.

4 Assumptions and limiting cases

The 47-point `knot_3.1.txt` geometry belongs to one KnotPlot-native family. Its numerical centerline has no authenticated Euler-equilibrium relaxation, no independently verified source-native topology and no SI radius. The original A064 centerline is placed in a periodic cubic domain of length $L = 2\pi$ after arbitrary normalization; Gaussian smoothing of its line vorticity uses $a = 0.4$ nominal units and $\Gamma = 1$ model unit. The finite-core filter is a numerical modeling assumption, *not* a derivation of canonical r_c. Actual 3D velocity and vorticity evolve without a driving frequency. A Hilbert-transform phase of a transient need not be a stable clock. A transported surface is material by construction but is a vortex tube only if $\boldsymbol{\omega}\cdot\mathbf{n}$ is suitably small on the surface. A low full-field relative return is a candidate, not a Newton–Krylov shooting certification.

5 Symbols, dimensions and units

Symbol	Meaning	SI dimensions if calibrated
$\mathbf{u}, \boldsymbol{\omega}$	velocity, curl of velocity	LT^{-1}, T^{-1}
L, a, Γ	box length, core width, circulation	L, L, L^2T^{-1}
E	mean energy per unit mass	L^2T^{-2}
H	mean helicity	LT^{-2}
Z	mean enstrophy	T^{-2}
C, λ_i	covariance and its eigenvalues	L^2
q_{sh}, q_Z	two dimensionless functionals	1
$\phi, \widehat{\Omega}$	observed phase, angular slope	$1, T^{-1}$
$\mathbf{X}, \mathbf{n}$	material nodes, unit normal	$L, 1$
T_R, g	relative return period, discrete/continuous symmetry	$T, 1$
M_g, μ	relative monodromy, multiplier	$1, 1$

No dimensional conversion to s, m or kg is made. In particular no SST circulation scale or relativistic time-dilation law is used to set a frequency.

6 Actual continuous and numerical model

The finite-dimensional Galerkin operator implements the projected Lamb form of Euler:

$$\partial_t \mathbf{u} = \mathcal{P}[\mathbf{u}\times\boldsymbol{\omega}], \quad \boldsymbol{\omega} = \nabla\times\mathbf{u}, \quad \nabla\cdot\mathbf{u} = 0. \tag{1}$$

The original A064 code deposits a signed line-current along each polygon edge (cloud-in-cell), applies Gaussian finite-core smoothing and a two-thirds wavenumber cutoff, then constructs the divergence-free Fourier Biot–Savart velocity

$$\widehat{\mathbf{u}}_0(\mathbf{k}) = \frac{i\mathbf{k}\times\widehat{\boldsymbol{\omega}}_a(\mathbf{k})}{|\mathbf{k}|^2} \quad (\mathbf{k}\neq 0). \tag{2}$$

RK4 evolves this field without inserted phase. Conservation screens use

$$E = \frac{1}{2V}\int_V |\mathbf{u}|^2\,dV, \quad H = \frac{1}{V}\int_V \mathbf{u}\cdot\boldsymbol{\omega}\,dV, \quad Z = \frac{1}{2V}\int_V |\boldsymbol{\omega}|^2\,dV. \tag{3}$$

For the full periodic volume, use enstrophy weights to construct the periodic-centered three-dimensional covariance tensor C. Ordered eigenvalues define the shape scalar

$$q_{\mathrm{sh}} = \frac{\lambda_1-\lambda_2}{\operatorname{tr} C}, \quad q_{\mathrm{sh},2} = \frac{\lambda_2-\lambda_3}{\operatorname{tr} C}. \tag{4}$$

These two eigenvalue gaps are correlated. Instead of treating them as independent physical clocks, use the distinct flow functional

$$q_Z = \frac{L^2 Z}{E}. \tag{5}$$

Both q_{sh} and q_Z are computed from each actual evolved 3D field, not from imposed oscillators. The analytic-signal phase estimator is

$$\phi_q(t) = \operatorname{unwrap}\arg\{[q(t)-\bar{q}] + i\mathcal{H}[q(t)-\bar{q}]\}, \qquad \widehat{\Omega}_q = \operatorname{slope}_W \phi_q(t). \tag{6}$$

This estimator is a screen only; it does not establish a relative periodic orbit or intrinsic clock. Thresholds were specified in method preflight: at least six unwrapped cycles; RMS phase-fit error at most 0.20 radians; instantaneous frequency CV at most 0.12; cross-frequency relative error at most 0.03. A constant shape under rigid rotation is no clock. A few cycles of a windowed transient are not a proof of stability. Time series are uniformly sampled by construction.

v0.4.4 material and stability operators

The finite-core tube seeded around the native trefoil is a *candidate* material surface. At every Euler RK4 substage its nodes obey

$$\frac{d\mathbf{X}}{dt} = \mathbf{u}(\mathbf{X},t), \quad \eta_\omega = \frac{1}{A}\int_S \frac{|\boldsymbol{\omega}\cdot\mathbf{n}|}{\max(|\boldsymbol{\omega}|,\epsilon_\omega)}\,dA, \quad \delta_V = \frac{|V(t)-V(0)|}{|V(0)|}. \tag{7}$$

The prescribed screening limits are $\delta_V < 0.05$, area-mean $\eta_\omega < 0.05$, and pointwise maximum $|\widehat{\omega}\cdot\mathbf{n}| < 0.10$, at all retained snapshots. Net vorticity flux through a closed surface is *not* an admissible vortex-tube test; it vanishes even on surfaces not made of vortex lines. Initial tube parameters are fixed at 32 axial sections, 10 azimuthal sides and radius 0.12 model units.

The full-field symmetry-reduced return statistic is

$$r_{\mathrm{RPO}}(T,g) = \frac{\|\mathbf{u}(T) - g\mathbf{u}(0)\|_{L^2(V)}}{\|\mathbf{u}(0)\|_{L^2(V)}}, \qquad g \in \mathbb{T}^3 \rtimes O^+_{\mathrm{cube}}. \tag{8}$$

The implementation registers a 0.025 relative candidate screen and searches all 24 proper cubic rotations with 3D periodic translations. This is a *full 3D-field search*, not proof of a converged nonlinear RPO. Repeated returns, neutral drift modes, shooting and mesh/time convergence remain necessary.

The independent tangent equation is evaluated using the unmodified A064 Euler linearization,

$$\dot{\delta\mathbf{u}} = \mathcal{P}[\delta\mathbf{u}\times\boldsymbol{\omega} + \mathbf{u}\times(\nabla\times\delta\mathbf{u})], \qquad M_g = g^{-1}D\Phi^{T_R}(\mathbf{u}_0), \tag{9}$$

where $M_g v = \mu v$, $\Omega_F = |\arg\mu|/T_R$ and $\epsilon_F = \|M_g v - \mu v\|/\|M_g v\|$. This is an *operator* frequency; a measured q-phase never enters the tangent propagator. Ritz values from a five-vector fixed-seed subspace are diagnostics only. No physical Floquet frequency can be reported without a validated RPO, residual, spectral convergence and a withheld prediction comparison.

Machine-readable science contract

Research question Does a material toroidal surface evolved synchronously with autonomous 3D Euler form an actual vortex tube, and does its full velocity field exhibit a certified relative periodic orbit with an independently calculated Euler tangent/Floquet frequency?

Objective Execute genuine Lagrangian material-surface RK4 tracking, require local vorticity tangency and volume transport, search all three velocity components for relative returns over the 24 proper cube rotations and periodic translations, validate the analytic Euler tangent linearization, and condition matrix-free monodromy on validated RPO. Physical PKLSA clock remains UNRESOLVED without convergence and distinct providers. Register and test both 2015 historical trefoil parametrizations (A Fourier p.32, B toroidal p.66) as independent *geometric

shapes*, not independent dynamics providers; verify exact A004 v0.2.2 SHA-256 provenance across 40 external samples and measure pilot Euler response with matched algorithmic parameters. v0.4.4 adds separately quantified material mesh/satellite deformation and vorticity-line candidates, matrix-free nonlinear full-field shooting and Floquet Arnoldi, and real optional Intel XPU 3D FFT/RK4 parity; uses v1.0.10 full diagnostic semantics without physical gate promotion. v0.4.4 adds separately registered T=20 three-mesh Lagrangian/Cauchy tests, sampled collision broadphase, multi-anchor vortex-line return map and Arc XPU environment attestation. These are diagnostic screens, not topology or frequency certification.

H_0 H0: current source-initialized Euler dynamics provides no qualified, reproducible material-vortex intrinsic clock; apparent Hilbert phase may be transient deformation, core/mesh artefact or spectral leakage.

H_1 H1: after independent material tube, relative periodic orbit and source qualification, the Euler/PKLSA vortex has a coherent recurrent phase stable under mesh, timestep and finite-core changes with a withheld Floquet frequency prediction.

Registered equations

E1: Unforced incompressible 3D Euler in projected Lamb form

$$\partial_t \mathbf{u} = \mathcal{P}[\mathbf{u} \times (\nabla \times \mathbf{u})], \qquad \nabla \cdot \mathbf{u} = 0 \tag{10}$$

Dimensional check: Both sides L T^-2; projection is dimensionless.

E2: Biot–Savart initialization from source-native line vorticity

$$\widehat{\mathbf{u}}(\mathbf{k}, 0) = \frac{i\mathbf{k} \times \widehat{\boldsymbol{\omega}}_a(\mathbf{k})}{|\mathbf{k}|^2}, \qquad \mathbf{k} \neq 0 \tag{11}$$

Dimensional check: i k cross omega_hat / k^2 gives velocity dimension L T^-1.

E3: Integral diagnostics energy, helicity, enstrophy

$$E = \frac{1}{2V}\int_V |\mathbf{u}|^2 dV, \quad H = \frac{1}{V}\int_V \mathbf{u} \cdot \boldsymbol{\omega}\, dV, \quad Z = \frac{1}{2V}\int_V |\boldsymbol{\omega}|^2 dV \tag{12}$$

Dimensional check: E: L^2 T^-2; H: L T^-2; Z: T^-2.

E4: Rotation-invariant shape observable from full 3D vorticity

$$q_{\mathrm{sh}}(t) = \frac{\lambda_1[C(t)] - \lambda_2[C(t)]}{\operatorname{tr} C(t)} \tag{13}$$

Dimensional check: Eigenvalue difference and covariance trace both L^2.

E5: Distinct dimensionless scalar flow functional

$$q_Z(t) = \frac{L^2 Z(t)}{E(t)} \tag{14}$$

Dimensional check: L^2 times T^-2 divided by L^2 T^-2 = 1.

E6: Signal-derived Hilbert phase and frequency without injection

$$\phi_q(t) = \operatorname{unwrap}\arg(q - \overline{q} + i\mathcal{H}[q - \overline{q}]), \qquad \widehat{\Omega}_q = \operatorname{slope}_{t\in W}\,\phi_q(t) \tag{15}$$

Dimensional check: q and phase dimensionless, Omega T^-1.

E7: Noncommuting temporal, spatial and core convergence probes

$$\delta_Z(A,B) = \frac{|Z_A(T) - Z_B(T)|}{\max(|Z_B(T)|, \epsilon_Z)} \tag{16}$$

Dimensional check: Ratio of two enstrophy magnitudes dimensionless; epsilon_Z must have same units as Z.

E8: Synchronous Lagrangian advection

$$\dot{\mathbf{X}} = \mathbf{u}(\mathbf{X}, t) \tag{17}$$

Dimensional check: Both sides L T^-1.

E9: Vortex-tube tangency and numerical material volume

$$\eta_\omega = \frac{1}{A}\int_S \frac{|\boldsymbol{\omega}\cdot\mathbf{n}|}{\max(|\boldsymbol{\omega}|, \varepsilon_\omega)}\, dA, \qquad \delta_V = \frac{|V(t) - V(0)|}{|V(0)|} \tag{18}$$

Dimensional check: Dimensionless ratios; excludes inference from net closed-surface flux.

E10: Full-field relative-periodic shooting residual

$$r_{\mathrm{RPO}}(T, g) = \frac{\|\mathbf{u}(T) - g\mathbf{u}(0)\|_{L^2(V)}}{\|\mathbf{u}(0)\|_{L^2(V)}}, \quad g \in \mathbb{T}^3 \rtimes O^+_{\mathrm{cube}} \tag{19}$$

Dimensional check: Same velocity dimensions cancel; T has dimensions of time.

E11: Euler tangent equation for independent monodromy

$$\dot{\delta\mathbf{u}} = \mathcal{P}[\delta\mathbf{u}\times\boldsymbol{\omega} + \mathbf{u}\times(\nabla\times\delta\mathbf{u})] \tag{20}$$

Dimensional check: Each side L T^-2.

E12: Relative monodromy and Floquet frequency

$$M_g = g^{-1}D\Phi^{T_R}(\mathbf{u}_0), \quad M_g v = \mu v, \quad \Omega_F = |\arg\mu|/T_R \tag{21}$$

Dimensional check: M and mu dimensionless; Omega_F dimension T^-1. Period cannot be fitted from scalar phase.

E13: Full-vector Ritz eigen-residual

$$\epsilon_F = \frac{\|M_g v - \mu v\|_{L^2}}{\|M_g v\|_{L^2}} \tag{22}$$

Dimensional check: Ratio dimensionless, computed on full 3D tangent field.

E14: Closed polyline arclength and sampling coefficient of variation

$$\ell_N = \sum_{j=0}^{N-1} \|\mathbf{X}_{j+1} - \mathbf{X}_j\|, \quad CV_e = \frac{\mathrm{sd}(\ell_j)}{\mathrm{mean}(\ell_j)}, \quad \mathbf{X}_N = \mathbf{X}_0 \tag{23}$$

Dimensional check: Arclength shares source coordinate units (arbitrary L); CV dimensionless.

E15: Centerline point-resolution sensitivity at fixed Euler field mesh

$$\delta_{Z,\mathrm{seed}} = \frac{|Z_{600}(T) - Z_{1200}(T)|}{\max(|Z_{1200}(T)|, \epsilon_Z)} \tag{24}$$

Dimensional check: Z and epsilon_Z share T^-2; delta dimensionless. This is NOT 3D mesh convergence.

E16: Cauchy deformation gradient

$$\dot{F} = (\nabla u)(X,t)F, \quad J = \det F \tag{25}$$

Dimensional check: F dimensionless, J dimensionless

E17: Material-mesh discretization convergence

$$\delta_{V,h} = |dV_{h/2} - dV_h| \tag{26}$$

Dimensional check: dimensionless

E18: Vorticity-line ODE and closure

$$dX/ds = \omega(X)/|\omega(X)|, \; c = |X(s = L) - X(0)| \tag{27}$$

Dimensional check: s and c have length units

E19: Nonlinear full-field relative shooting

$$F(U_0, T, g) = g^{-1}\Phi_T(U_0) - U_0 \tag{28}$$

Dimensional check: velocity dimensions

E20: Second independent relative return

$$r_2 = \|g^{-2}\Phi_{2T}(U_0) - U_0\|/\|U_0\| \tag{29}$$

Dimensional check: dimensionless

E21: Independent Arnoldi monodromy

$$M_g v = \mu v,\ \epsilon = \|M_g v - \mu v\|/\|M_g v\| \tag{30}$$

Dimensional check: dimensionless

E22: Independent XPU/FP64 solver parity

$$\delta_{gpu} = \|u_{xpu}(T) - u_{fp64}(T)\|_2/\|u_{fp64}(T)\|_2 \tag{31}$$

Dimensional check: dimensionless

E23: Long-horizon volume error across three independently advected mesh resolutions

$$\delta_V^{(h)}(t) = |V_h(t) - V_h(0)|/|V_h(0)| \tag{32}$$

Dimensional check: dimensionless

E24: Eulerian-to-Lagrangian Cauchy deformation determinant screen

$$J(t) = \det(\partial X(t)/\partial X(0)) \tag{33}$$

Dimensional check: dimensionless

E25: Vorticity streamline return map without endpoint stitching

$$d_{\text{close}} = \min_{m\in\mathbb{Z}^3} \|X(L) - X(0) + 2\pi m\| \tag{34}$$

Dimensional check: length in model units

E26: Hardware XPU whole-field RK4 parity statistic

$$\epsilon_{\text{XPU}} = \|u_{\text{XPU}}(T) - u_{\text{CPU}}(T)\|_2/\|u_{\text{CPU}}(T)\|_2 \tag{35}$$

Dimensional check: dimensionless

Registered code-to-mathematics steps

1. **S1** [G1; equations E2]: Hash 6 source files and exact A064 solver copy, reject mutations
2. **S2** [G1; equations E6]: Unblind method controls positive oscillator, rigid rotation and beating; preserve as method-only
3. **S3** [G2; equations E2]: Normalize unchanged native XYZ into periodic 2pi cube; deposit signed circulation; apply Gaussian finite-core and transverse Leray projection
4. **S4** [G2; equations E1, E3]: Run genuine unforced spectral 3D Euler/RK4 trajectory to T=20, n16 dt.04 with 251 snapshots; compute energy/helicity/CFL/divergence
5. **S5** [G3; equations E4, E5, E6]: Compute independent functionals qshape and qZ from evolved 3D fields then Hilbert phase, no oscillator injected

6. **S6** [G5; equations E1, E7]: Rerun n24, n16 dt=.02 and two finite-core branches (.32,.48) to fixed T, report differences even when science gate G5 is blocked

7. **S7** [G3; equations E4, E6]: Do not mark G3 terminal without qualified material tube and relative periodic orbit; do not turn unresolved into fail

8. **S8** [G4; equations E1, E6]: Register future independent Euler/Floquet frequency on a withheld provider before running G4-G8

9. **S9** [G2; equations E8, E9]: Create a closed toroidal polygon mesh from original native trefoil, pre-register radius, sections and sides; keep separate from vortex qualification.

10. **S10** [G3; equations E1, E8, E9]: Evolve material mesh nodes with RK4 positions synchronized to same Euler intermediate stages and evaluate volume and omega-normal diagnostics at all snapshots.

11. **S11** [G3; equations E10]: Search full 3-component Euler velocity-field return under translations and proper cubic rotations; report minimum residual without scalar phase fit.

12. **S12** [G2; equations E11, E12]: Independently compare analytic Euler tangent derivative to centered finite difference and validate manufactured two-mode Floquet operator oracle.

13. **S13** [G4; equations E11, E12, E13]: Only on RPO plus tube screening PASS calculate a finite low-rank relative monodromy Ritz screen with full-state residuals; never claim convergence/physics without shooting and grid/provider gates.

14. **S14** [G1; equations E14]: G1: Validate both 2015 A/B historical formula-manifest families and all 40 byte-exact XYZ files; confirm orientation/reversal/reflection transforms and record outer-vs-inner version discrepancy.

15. **S15** [G2; equations E1, E2, E3, E15]: G2 diagnostic sidecar: evolve A/B N600 seeds in BASIC and registered N600/N1200/reflection/reversal controls in FULL/CERTIFY with unchanged A064 Euler 3D at fixed model parameters; never count controls as independent physical providers. In FULL/CERTIFY, the A and B N600 runs also track transported material surfaces, perform identity-rotation full-vector endpoint translation screens (not full 24-rotation RPO shooting), and check analytic tangent Euler via finite differences; no physical Floquet frequency or RPO certification.

16. **S16** [G3; equations E8, E9, E16, E17]: Track three material mesh resolutions simultaneously with exact Euler RK4 stages and Cauchy satellite tracers

17. **S17** [G3; equations E18, E9]: Construct candidate vortex surface by integrating actual Euler vorticity field lines and rejecting nonclosure

18. **S18** [G3; equations E19, E20]: For qualified return candidates attempt matrix-free nonlinear shooting and reintegrated second-return closure, without fitting any phase frequency

19. **S19** [G4; equations E21, E12]: Given eligible candidates, calculate independent matrix-free Arnoldi Floquet multipliers and full-field eigen-residuals

20. **S20** [G6; equations E22, E1]: Run genuine 3D Euler torch-XPU FFT/RK4 parity to NumPy FP64, prohibit implicit CPU fallback; require independent official framework SYCL/DD32 checks

21. **S21** [G8; equations E16, E21, E22]: Apply Framework 1.0.10 FULL_DIAGNOSTIC run_evaluators even if upstream scientific screen FAIL; separate raw status from recursive scientific authority

22. **S22** [G5; equations E23, E24]: Compute registered T=20 3-mesh Lagrangian transport with finite-sampled material volume and detF, never certify global embeddedness

23. **S23** [G3; equations E25]: Scan preregistered vorticity-return anchors and lengths; record closure without stitching or pseudo-replication

24. **S24** [G6; equations E26]: Record Arc XPU environment; execute same-seed GPU FP32/FP64 field parity where actual hardware exists, with no fallback

7 Executable algorithm and fixed run matrix

1. Verify byte-exact historical controls and A064 solver/source hashes; refuse changed or missing sources.

2. Load 47 native 3D XYZ vertices; normalize and map to the periodic box; deposit vorticity and compute solenoidal $\mathbf{u}(0)$ from A064.

3. In `BASIC`, evolve $N = 16$, $a = 0.40$, $\Delta t = 0.04$, $T = 6$, retaining 151 genuine 3D field snapshots and advected material mesh states.

4. In `FULL/CERTIFY`, evolve $N = 16$, $a = 0.40$, $\Delta t = 0.04$, $T = 20$, retaining 251 3D fields and synchronously advected material surfaces. Independently evolve $N = 24$ at identical $(a, \Delta t, T)$, $N = 16$ at $\Delta t = 0.02$, and $N = 16$ with $a = 0.32, 0.48$ at fixed $(T, \Delta t)$.

5. At every retained field, measure material-volume drift, surface vorticity tangency, and E, H, Z, divergence, CFL, shape tensor, q_{sh} and q_Z. Recover phase only from measured functionals; do not input a theoretical frequency.

6. Evaluate complete Euler state returns up to the 24 cubic rotations and 3D translations, retaining top RPO candidates and relative L^2 residues. Validate tangent-linear Euler against finite differences and a manufactured analytic two-mode Floquet oracle. Only attempt matrix-free Ritz on admissible candidates.

7. Require numerical conservation to close G2; record spatial/time/core diagnostic differences even when downstream physical gates are blocked. Do not promote G3 unless material-tube tangency, a verified nonlinear relative periodic orbit, independent providers and independently predicted frequency all qualify.

8. Let the official v1.0.10 runtime write unique BLIND ZIPs, PDFs and gate-ledger hashes; do not bypass reveal policy.

8 Source provenance and provider independence

The 40 historical XYZ curves are imported exclusively from the externally installed, SHA-256-pinned `A004_historical_trefoil/SST_Historical_Trefoil_XYZ_v0.2.2/data` release under `02_libraries/A_knot_libraries`. There are no internal A068 copies or automatic fallback; missing or changed A004 data fails G1 closed. The orientation reversal holds the first vertex fixed. Historical A/B shapes and controls do not provide independent physical PKLSA dynamical evidence.

`knot_3.1.txt`: SHA-256 `98a628af4ba5d09274f0a1845c351a7d633a39ff833a3ae0b78e351bafdabf`. The original A064 Python Euler solver is included byte-for-byte at `experiment/euler3d.py` and as historical reference. Earlier two A068 preflight JSONs and A064-v0.3.0 results are immutable controls, not independent vortex providers. The finite-core smoothing function, Fourier truncation and normalization are inherited from A064. The expected upstream hashes are enforced in G1, and mesh/core variation does not create distinct provider families. Input XYZ geometry has *unverified* physical vortex equilibrium and no SI units. A tracer torus, RPO scan, or tangent derivative constitutes a distinct numerical operator, not a distinct physical provider.

v0.4.4: Full diagnostic program and GPU authority

This instance uses Framework v1.0.10 as a **LOCAL_VALIDATED_CANDIDATE**, not canonical hardware certification. Independently executable scientific gates use FULL_DIAGNOSTIC scheduling: an earlier numerical FAIL cannot erase later measured diagnostic results, but downstream claims remain DIAGNOSTIC_ONLY when prerequisites are unqualified. No REVEAL is permitted without an independently qualified final gate.

Tranche M: The same Euler RK4 stage states advect three material meshes (32×10, 64×20, 128×40). A separate set of nearby tracers estimates $F_{ij} = \partial X_i / \partial X_{0j}$ and $\det F$. Mesh degeneracy is reported, while self-intersection/embeddedness remains unverified.

Tranche V: Candidate vortex surfaces are generated by integrating $dX/ds = \omega/|\omega|$ from transverse seed rings. End-to-start closure and local $\omega \cdot n$ are measured: failed closure is not repaired by forcing a seam.

Tranche R: Full-field RPO candidates may enter fixed-group fixed-period Newton–GMRES shooting, a second independently reintegrated return and symmetry-residual measurements. Optimizing the free period and group phase, neutral-mode gauge fixing and spatial convergence remain independent certification requirements.

Tranche F: The Euler tangent propagator and matrix-free Arnoldi eigenpairs are independent of the measured scalar phase. Real Floquet frequencies are strictly conditional on an admissible RPO; the manufactured two-mode eigenvalue is a software test only.

Tranche GPU: A genuine three-dimensional FFT/RK4 solver using PyTorch XPU on Intel Arc runs as a separate `float32` screening lane and conditional `float64` parity lane. Every run must record actual XPU device, full-state L2 error against independently integrated NumPy FP64, energy and helicity differences. Missing XPU gives `UNAVAILABLE_NO_FALLBACK`, never a CPU substitute. Unsupported FP64 is separately recorded while FP32 remains screening only. Framework SYCL FP32/DD32 common-kernel parity is *separate* and must be hardware-tested on the target Arc A770 before v1.0.10 canonical promotion.

Registered cost bound: BASIC mesh deformation $T = 0.24$, FULL $T = 0.80$, CERTIFY $T = 0.80$ with halved step; main A068 Euler campaigns retain their longer trajectories. These mesh studies are numerical prescreens, not full physical time convergence. All 40 external A004 v0.2.2 XYZ hashes and older A064 files remain unchanged.

Additional registered operators (E16–E22).

$$\dot{F} = (\nabla \mathbf{u})F, \qquad J = \det F, \tag{36}$$

$$\delta_{V,h} = |\delta V_{h/2} - \delta V_h|, \tag{37}$$

$$\frac{d\mathbf{X}}{ds} = \frac{\boldsymbol{\omega}(\mathbf{X})}{|\boldsymbol{\omega}(\mathbf{X})|}, \tag{38}$$

$$\mathbf{R}(U_0, T, g) = g^{-1}\Phi_T(U_0) - U_0, \tag{39}$$

$$r_2 = \frac{\|g^{-2}\Phi_{2T}(U_0) - U_0\|}{\|U_0\|}, \tag{40}$$

$$M_g \mathbf{v} = \mu \mathbf{v}, \qquad \epsilon_F = \frac{\|M_g \mathbf{v} - \mu \mathbf{v}\|}{\|M_g \mathbf{v}\|}, \tag{41}$$

$$\delta_{\mathrm{GPU}} = \frac{\|\mathbf{u}_{\mathrm{XPU}}(T) - \mathbf{u}_{\mathrm{FP64}}(T)\|_2}{\|\mathbf{u}_{\mathrm{FP64}}(T)\|_2}. \tag{42}$$

All ratios are dimensionless. F is an operational deformation-gradient estimate, not an independently reconstructed physical vortex equilibrium. M_g is conditional on a qualified relative return; no observed Hilbert frequency enters the operator.

Historical A/B trefoil geometric seed atlas (new v0.4.4)

Two distinct 2015 notebook-parametric trefoils are provided as headerless, byte-preserved XYZ arrays, not as physical PKLSA vortex equilibria:

$$\mathbf{X}_A(t) = \big(\cos t + 1.5\cos 2t,\ \sin t - 1.5\sin 2t,\ 0.35\sin 3t\big), \tag{43}$$

$$\mathbf{X}_B(t) = \big((2+\cos 3t)\cos 2t,\ (2+\cos 3t)\sin 2t,\ \sin 3t\big), \quad t \in [0, 2\pi). \tag{44}$$

Analytic reference lengths are $\ell_A = 19.976521990340952$ and $\ell_B = 31.898600666412335$ historical coordinate units. Each has $N \in \{600, 1200, 2400, 4800, 9600\}$ with original/reversed and X-mirrored/reversed controls: $2 \times 5 \times 4 = 40$ immutable files. Each vertex file has implicit periodic closure; the true equal-arc-length sampling and SHA-256 are checked against the A004 v0.2.2 manifest. Reversal of polygon traversal reverses the imposed line-current direction in the inherited Euler initializer; $x \mapsto -x$ changes chirality but is not an independent physical realization.

$$\ell_N = \sum_{j=0}^{N-1} \|\mathbf{X}_{j+1} - \mathbf{X}_j\|, \quad CV_e = \frac{\mathrm{sd}(\ell_j)}{\mathrm{mean}(\ell_j)}, \qquad \mathbf{X}_N = \mathbf{X}_0. \tag{45}$$

At fixed Galerkin grid size n, core a, Δt and T, $\delta_{Z,\mathrm{seed}} = |Z_{N600}(T) - Z_{N1200}(T)| / \max(|Z_{N1200}(T)|, \epsilon_Z)$ measures *source-centerline sampling* error. It is explicitly not Euler grid convergence. BASIC runs two short A/B geometry-initiated 3D Euler pilots; FULL/CERTIFY separately run eight preregistered A/B sampling, mirror and traversal controls. For the A/B original N600 candidates in FULL/CERTIFY, an advected toroidal surface, full-vector endpoint translation screen (identity rotation only) and separately finite-difference-verified Euler tangent operator are also computed. This historical sidecar does *not* perform 24-rotation RPO shooting or a certified Floquet prediction; those are distinct primary A068 operators and require qualified physical inputs. All original XYZs are dimensionless, non-SI and geometrical only. **Pinned A004 source:** the preceding v0.4.1 development archive reported an outer-v0.2.0 / inner-v0.1.0 provenance inconsistency. The present v0.4.4 release uses instead the separately verified A004 `v0.2.2` manifest (SHA-256 `e7477a9c817f59124f279d08f001f5ef7f85c0a7c178e0ecf5b0b0116e57083e`) and all 40 external XYZ hashes. Its `cw` ordering preserves the first vertex. A/B geometry and controls contribute **zero independently qualified physical PKLSA providers** and cannot independently close G3–G8.

9 Dependency graph and physical guardrails

Gate	Required decision	Initial status
G0	Framework freeze, signed provenance path	runtime
G1	Native geometry, solver and historical control hashes	source check
G2	Real Euler 3D evolution plus $E, H, \nabla \cdot \mathbf{u}$, CFL	numerical PASS/FAIL
G3	Six clean cycles; *material tube and RPO independently qualified*	UNRESOLVED
G4	Independent withheld Euler/Floquet prediction	blocked
G5	Converged space, time and core with physical state fixed	blocked
G6	Explicit native backend parity/authority	blocked
G7	Multiple independent qualified PKLSA providers	blocked
G8	Physical mechanism conclusion if all required gates pass	blocked
G9	Nonced reveal commitments on allowed frozen gate ledger	withheld

Gate order is an explicit DAG. A negative numerical screen does not count as a physical hypothesis falsification. A gapped clock phase without an identified finite-core material vortex can *never* close G3 in this release.

10 Numerical control and exploratory observations

For G2 preregistered development thresholds: relative divergence $< 10^{-9}$; relative energy drift $< 10^{-5}$; relative helicity drift $< 10^{-4}$; CFL < 0.25; finite velocity field, and nontrivial evolution. Comparative tests use

$$\delta_Z(A,B) = \frac{|Z_A(T) - Z_B(T)|}{\max(|Z_B(T)|, 10^{-30})}. \tag{46}$$

Exploratory, **unblinded** local FULL prechecks (not an official frozen framework run):

Case	Grid, core, Δt	$Z(T)$	$\lvert\Delta E/E\rvert_{\max}$	Phase cycles
Base	$16, 0.40, 0.04$	0.064202	8.34×10^{-10}	0.646
Spatial	$24, 0.40, 0.04$	0.121697	6.93×10^{-9}	0.615
Temporal	$16, 0.40, 0.02$	0.064202	4.22×10^{-11}	0.646
Core low	$16, 0.32, 0.04$	0.094750	1.69×10^{-9}	0.643
Core high	$16, 0.48, 0.04$	0.035642	2.04×10^{-10}	0.723

δ_Z under time-step halving is 2.01×10^{-9}, while spatial $\delta_Z \simeq 0.472$ and core-spread $\simeq 0.921$. Thus temporal integration control succeeds but spatial/core physical convergence demonstrably **does not** qualify. Less than one observed phase cycle fails the method screen's six-cycle minimum; it cannot be extrapolated to a period by assumption. These values are preliminary numerical model diagnostics and will be recomputed on each framework run.

11 Backend authority

The primary reference field integration is the upstream NumPy/SciPy FP64 Euler simulator. An optional independent torch-XPU full 3D Euler lane reports actual device and field parity; its presence does not certify a physical vortex. Its runs are not equivalent to native C++ FP64 or Intel SYCL parity certification. Framework v1.0.10 backend configuration may disable certain gates; disabled scientific gates do not pass. DD32/FP32 only screen, never physical confirmation. All core/mesh branches here share one upstream mathematical method and source provenance.

12 Blindness, frozen protocol and reveal

The historical source and this exploratory run have already been inspected, so only framework *output confidentiality and audit integrity* are blind. Future provider-independent PKLSA-clock claims must be preregistered on a fresh held-out source before opening it. Frozen v1.0.10 publishes run-specific ZIP/PDF identities and archives previous output roots; `REVEAL_IF_ALLOWED` correctly refuses when a required physical G8 is diagnostically `UNRESOLVED`. Never edit a frozen policy, ledger or HMAC commitment to force a reveal.

13 Statistics and confidence limits

The Euler trajectory is one deterministic single-provider numerical realization. Time snapshots, mesh steps and smoothing variants are *not* independent experimental replicates. A Hilbert phase slope on less than one cycle has no established asymptotic period. No physical confidence interval or p-value is computed. Discovery windows and candidate selection are fixed by code; no frequency target, SST phase law or operator eigenvalue is supplied.

14 Actual runtime results and retained evidence

Each official BASIC/FULL/CERTIFY pipeline computes and stores `EULER_DYNAMICS_RESULTS.json`, `EULER_3D_FIELDS_MODEL_ONLY.npz`, `MATERIAL_MESH_CONVERGENCE_AND_CAUCHY.json`, `LONG_`

HORIZON_MATERIAL_TUBE_V044.json, VORTEX_LINE_RETURN_MAP_V044.json, GPU_XPU_ENVIRONMENT_PREFLIGHT_V044.json, VORTEX_LINE_SEED_SCREEN.json, NONLINEAR_RPO_SHOOTING_AND_SECOND_RETURN.json, INDEPENDENT_FLOQUET_ARNOLDI.json, and GPU_XPU_FULL_EULER_RK4_PARITY.json. The HISTORICAL_2015_EULER_MODEL_DIAGNOSTICS.json remains a geometric-seed sidecar. The official framework writes hashed source/run manifests, a gate ledger, PDF and BLIND ZIP for each run. A BASIC result cannot substitute for FULL Euler mesh/core convergence. No GPU or physical-clock conclusion follows from an absent XPU device or an operator oracle alone.

Automatic run provenance

Field	Recorded value
Run ID	20261019T234856589838Z-b3612aeb
Run phase	BLIND
Protocol bundle SHA-256 (byte exact)	3df609eada0a9494cbb554930bb787746df77f663cc0e205 888a56fffab9c308
Portable protocol diagnostic SHA-256	3f9a033215efaad6175b8ef2b8f3d725ac0d37cf8b294675 c06c312f79765417
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Science evidence SHA-256	0d253d787041193a085d940f23feadd57af067bef68a0c85 62b40760e231befd
Final output manifest	See OUTPUT_MANIFEST.json (finalized after PDF and report metadata)

Gate ledger

Gate	Status	Question	Reason
G0	PASS	Did pinned framework 1.0.10 validate frozen A068 v0.4.4 science protocol (framework itself remains candidate until Windows/Arc validation)?	Official version-pinned v1.0.10 local candidate runtime.
G1	PASS	Are A064/KnotPlot provenance hashes and all 40 historical 2015 A/B XYZ source hashes and exact symmetry controls valid?	A004 external 40/40 provenance and exact frozen science operators
G2	PASS	Do full unforced 3D Euler model trajectories satisfy numerical conservation, including independent A004 geometric sidecars?	Real geometry-seeded 3D Euler RK4 simulation, no physical PKLSA authority
G3	UNRESOLVED	Do material mesh refinement, Cauchy det(F), vortex-line closure, full-field recurrent state and nonlinear shooting jointly qualify a physical vortex clock?	Independent material/RPO diagnostics executed; physical source, embedded material tube, full shooting and recurrence unqualified
G4	UNRESOLVED	Do independent Arnoldi Euler monodromy multipliers predict withheld observed frequency with converged full-field eigen residual?	Analytic oracle and tangent operator are diagnostics; no qualified RPO eigenfrequency for held-out comparison

G5	FAIL	Are mesh, timestep and core diagnostics converged independently even if earlier physics gates are negative?	Independent numerical convergence screen; v1.0.10 qualification remains DIAGNOSTIC_ONLY if G3/G4 unresolved
G6	UNRESOLVED	Does actual Arc XPU whole-solver FFT/RK4 pass NumPy FP64 parity without CPU fallback and separate framework SYCL checks?	No real Intel Arc/XPU runtime available; no CPU substitution
G7	UNRESOLVED	Are qualified clock results reproduced by truly independent PKLSA dynamics providers?	A004 trefoil geometry variants are not independent calibrated material vortex providers
G8	UNRESOLVED	Are all physical mechanism preconditions qualified for a terminal verdict?	No physical SST clock inference; no bypass of reveal eligibility
G9	DEFERRED	Reveal verification after registered blind physical conclusion.	Reveal is a separate, commitment-verified phase and is not executed during blind runs.

Scientific qualification (separate from measured gate outcomes)

Gate	Evidence qualification	Blocking prerequisites
G0	QUALIFIED	none
G1	QUALIFIED	none
G2	QUALIFIED	none
G3	UNRESOLVED	none
G4	DIAGNOSTIC_ONLY	G3
G5	DIAGNOSTIC_ONLY	G4
G6	DIAGNOSTIC_ONLY	G5
G7	DIAGNOSTIC_ONLY	G6
G8	DIAGNOSTIC_ONLY	G7
G9	DEFERRED	G8

Backend manifest


```
{
  "A004_external_pinned": true,
  "a004_version": "v0.2.2",
  "actual_backend": "python-numpy-fp64",
  "authority": "EULER_MODEL_SIMULATION_ONLY",
  "field_line_seed_diagnostic": true,
  "floquet": {
    "actual_euler_eigenfrequency": "UNRESOLVED_NO_CERTIFIED_RPO",
    "manufactured_oracle": {
      "omega_calculated": 1.7500000000000002,
      "omega_expected": 1.75,
      "oracle_pass": true
    }
  },
  "framework": "1.0.10_LOCAL_VALIDATED_CANDIDATE",
  "full_3d_trajectory_file": "EULER_3D_FIELDS_MODEL_ONLY.npz",
  "gpu": {
    "actual_backend": "UNAVAILABLE",
    "fp32_screen": {
      "actual_backend": "UNAVAILABLE",
      "environment": {
        "torch": "2.12.1+cpu",
        "xpu_available": false,
        "xpu_count": 0
      },
```

```
      "error": "RuntimeError: Intel XPU is not available; refusing CPU
fallback",
      "hardware_gpu_executed": false,
      "parity_pass": false,
      "requested_device": "xpu",
      "schema": "A068-GPU-TRIPLE-PARITY-v0.4.4",
      "status": "UNAVAILABLE_NO_FALLBACK"
    },
    "fp32_screen_pass": false,
    "fp64_lane": {
      "actual_backend": "UNAVAILABLE",
      "environment": {
        "torch": "2.12.1+cpu",
        "xpu_available": false,
        "xpu_count": 0
      },
      "error": "RuntimeError: Intel XPU is not available; refusing CPU
fallback",
      "hardware_gpu_executed": false,
      "parity_pass": false,
      "requested_device": "xpu",
      "schema": "A068-GPU-TRIPLE-PARITY-v0.4.4",
      "status": "UNAVAILABLE_NO_FALLBACK"
    },
    "fp64_parity_pass": false,
    "hardware_gpu_executed": false,
    "qualification":
"FP32_SCREENING_ONLY_FP64_REQUIRES_HARDWARE_PARITY_AND_FRAMEWORK_TESTS",
    "schema": "A068-GPU-2LANE-v0.4.4",
    "status": "UNAVAILABLE_NO_FALLBACK"
  },
  "gpu_environment_preflight": {
    "diagnostic": "Torch wheel, Intel GPU runtime or device unavailable;
check wheel, drivers and XPU support in this Python environment.",
    "environment_changes_performed": false,
    "no_cpu_fallback": true,
    "platform": "Windows-11-10.0.26300-SP0",
    "python": "3.14.3",
    "requested_backend": "pytorch-xpu",
    "schema": "A068-XPU-PREFLIGHT-v0.4.4",
    "status": "UNAVAILABLE_NO_FALLBACK",
    "torch_cuda_version": null,
    "torch_version": "2.12.1+cpu",
    "torch_xpu_available": false,
    "xpu_count": 0
  },
  "independent_physical_providers": 0,
  "long_material_diagnostic": true,
  "material_mesh_diagnostic": true,
  "no_native_cpp_or_sycl_dd32_whole_solver_certification_claim": true,
  "physical_intrinsic_clock": "UNRESOLVED",
  "schema": "SST-BACKEND-MANIFEST-2",
  "science_ledger_execution_policy": "FULL_DIAGNOSTIC_FAIL_NOT_STOP",
  "shooting": {
    "rpo_candidate_residual": 0.13378626193703955,
    "status": "NOT_ATTEMPTED_FULL_FIELD_CANDIDATE_SCREEN_FAILED"
  },
  "version": "A068-v0.4.4",
  "vortex_return_map_diagnostic": true
}
```


Source manifest


```
{
  "manifest_sha256":
```

```
"eb2f37acb45b22cfd68319239c14ff2232fb17d83f759028be8ac08b4425f4d3",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "A068",
      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "A064_DERIVED_SINGLE_PROVIDER",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifi
ers\\A068_PKLSA_PhaseClock\\A068-v0.4.4\\data\\PREFLIGHT_RESULTS.json",
      "provenance_family": "A064_KNOTPLOT_NATIVE_AND_PRE-FLIGHT",
      "registry_hint": {
        "catalog_id": "A068",
        "hits": []
      },
      "role": "historical_control",
      "sha256":
"eb2c099968b5ab2b33abb9ce855b0a4c77f58cd0f0137ccabdaf917146574730",
      "source_id": "S0",
      "version": "A068-v0.1.0_historical_unblinded"
    },
    {
      "catalog_id": "A068",
      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "A064_DERIVED_SINGLE_PROVIDER",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifi
ers\\A068_PKLSA_PhaseClock\\A068-v0.4.4\\data\\PREFLIGHT_WITH_A064.json",
      "provenance_family": "A064_KNOTPLOT_NATIVE_AND_PRE-FLIGHT",
      "registry_hint": {
        "catalog_id": "A068",
        "hits": []
      },
      "role": "historical_control",
      "sha256":
"7f6f865af5e6c459462bcc88780415a1d7257e96558cc47cae9ac6bd723f6902",
      "source_id": "S1",
      "version": "A068-v0.1.0_historical_unblinded"
    },
    {
      "catalog_id": "A068",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "A064_DERIVED_SINGLE_PROVIDER",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifi
ers\\A068_PKLSA_PhaseClock\\A068-v0.4.4\\data\\A064-v0.3.0_results.json",
      "provenance_family": "A064_KNOTPLOT_NATIVE_AND_PRE-FLIGHT",
      "registry_hint": {
        "catalog_id": "A068",
        "hits": []
      },
      "role": "solver_provenance",
      "sha256":
"3c1469cfddeafbc69c368a779072144119769d5ee9276a92064c803fbc6af3c1",
      "source_id": "S2",
      "version": "A064-v0.3.0"
    },
    {
      "catalog_id": "A068",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "KNOTPLOT_NATIVE_V012",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifi
ers\\A068_PKLSA_PhaseClock\\A068-v0.4.4\\data\\knot_3.1.txt",
      "provenance_family": "A064_KNOTPLOT_NATIVE_AND_PRE-FLIGHT",
```

```
      "registry_hint": {
        "catalog_id": "A068",
        "hits": []
      },
      "role": "seed_geometry",
      "sha256":
"98a628af4ba5d09274f0a1845c351a7a3d633a39ff833a3ae0b78e351bafdabf",
      "source_id": "S3",
      "version": "v0.4.4"
    },
    {
      "catalog_id": "A068",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "A064_DERIVED_SINGLE_PROVIDER",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifi
ers\\A068_PKLSA_PhaseClock\\A068-v0.4.4\\data\\A064_euler_tangent_upstream.py
",
      "provenance_family": "A064_KNOTPLOT_NATIVE_AND_PRE-FLIGHT",
      "registry_hint": {
        "catalog_id": "A068",
        "hits": []
      },
      "role": "solver_provenance",
      "sha256":
"a20863ba43277656dde1a638653aac5d4875d3880fef1ca930e6096de30b3621",
      "source_id": "S4",
      "version": "A064-v0.3.0"
    },
    {
      "catalog_id": "A068",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "A064_DERIVED_SINGLE_PROVIDER",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifi
ers\\A068_PKLSA_PhaseClock\\A068-v0.4.4\\data\\EULER_SOURCE_AUDIT.json",
      "provenance_family": "A064_KNOTPLOT_NATIVE_AND_PRE-FLIGHT",
      "registry_hint": {
        "catalog_id": "A068",
        "hits": []
      },
      "role": "solver_provenance",
      "sha256":
"95147f5e6b831ae9fd95d4f329b6ce832ce2d72bb74cc940cc12d87b38f49623",
      "source_id": "S5",
      "version": "A064-v0.3.0"
    },
    {
      "catalog_id": "A004",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family":
"A004_2015_TWO_GEOMETRIC_FAMILIES_NO_PHYSICAL_PROVIDER",
      "path": "C:\\workspace\\projects\\SST-Workbench\\02_libraries\\A_knot_l
ibraries\\A004_historical_trefoil\\SST_Historical_Trefoil_XYZ_v0.2.2\\data\\m
anifest.json",
      "provenance_family": "HISTORICAL_2015_NOTEBOOK",
      "registry_hint": {
        "catalog_id": "A004",
        "hits": []
      },
      "role": "external_A004_historical_geometry_manifest",
      "sha256":
"e7477a9c817f59124f279d08f001f5ef7f85c0a7c178e0ecf5b0b0116e57083e",
      "source_id": "S6",
```

```
      "version": "SST_Historical_Trefoil_XYZ_v0.2.2_PINNED"
    },
    {
      "catalog_id": "A004",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family":
"A004_2015_TWO_GEOMETRIC_FAMILIES_NO_PHYSICAL_PROVIDER",
      "path": "C:\\workspace\\projects\\SST-Workbench\\02_libraries\\A_knot_l
ibraries\\A004_historical_trefoil\\SST_Historical_Trefoil_XYZ_v0.2.2\\data\\A
_Fourier_2015_p32_orig_ccw_N600_arclength.xyz",
      "provenance_family": "HISTORICAL_2015_NOTEBOOK",
      "registry_hint": {
        "catalog_id": "A004",
        "hits": []
      },
      "role": "external_A004_historical_geometry",
      "sha256":
"8213fb6aa605adcebd45e05dca4a63abf2b94a204c47c5cdb0ca190794523f1e",
      "source_id": "S7",
      "version": "SST_Historical_Trefoil_XYZ_v0.2.2_PINNED"
    },
    {
      "catalog_id": "A004",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family":
"A004_2015_TWO_GEOMETRIC_FAMILIES_NO_PHYSICAL_PROVIDER",
      "path": "C:\\workspace\\projects\\SST-Workbench\\02_libraries\\A_knot_l
ibraries\\A004_historical_trefoil\\SST_Historical_Trefoil_XYZ_v0.2.2\\data\\B
_Torus_2015_p66_orig_ccw_N600_arclength.xyz",
      "provenance_family": "HISTORICAL_2015_NOTEBOOK",
      "registry_hint": {
        "catalog_id": "A004",
        "hits": []
      },
      "role": "external_A004_historical_geometry",
      "sha256":
"ace1f76cb082639b31515c772e39141aef6ede2781371e1402f1019652f9b7ab",
      "source_id": "S8",
      "version": "SST_Historical_Trefoil_XYZ_v0.2.2_PINNED"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```


15 Interpretation and next falsification step

The existence of nonlinear Euler evolution does not demonstrate a stable toroidal material boundary, spectral convergence, global RPO or an intrinsic physical clock. Future v0.4 qualification needs a source-native independent core field, advected material boundary with demonstrated low flux, full-state recurrence modulo allowed symmetries, a separately computed time-dependent Floquet/Koopman prediction and a new source/protocol commitment. Historical A064 spectral eigenpairs at $N = 8, 12, 16$ were nonconvergent and cannot be used as the predicted clock frequency here. Likewise, no comparison to Lorentz time dilation is permitted before a physical intrinsic oscillator is certified.

16 Reproducibility and bibliography

Install only into a *new* A068-v0.4.4 instance with the official `tools/new_falsifier.py` from locally present v1.0.10. Run `pytest -q tests`, then `run_all.cmd FREEZE`, `run_all.cmd BASIC`,

`run_all.cmd FULL`, and optionally `run_all.cmd CERTIFY`; no frozen package from v0.2.0 is overwritten. Preserve all v0.2.0 result evidence and commits.

References

[1] P. G. Saffman, *Vortex Dynamics* (Cambridge University Press, 1992), doi:10.1017/CBO9780511624063.

[2] H. K. Moffatt, J. Fluid Mech. **35**, 117–129 (1969), doi:10.1017/S0022112069000991.

[3] D. Gabor, J. Inst. Elect. Eng. Part III **93**, 429–457 (1946), doi:10.1049/ji-3-2.1946.0074.

17 v0.4.4 registered long-horizon diagnostic addendum

The FULL profile advances all three registered toroidal material meshes using identical Euler–RK4 stages through $T = 20$ model time. The independently computed volume histories $\delta_V^{(h)}$ and satellite $\det F$ are screened at a finite set of snapshots. A sampled triangle contact broadphase may identify potential mesh intersections but cannot prove embeddedness. The registered multi-anchor vorticity return map reports genuine integral curve endpoint errors before any closure. GPU measurements use the actual XPU device, if available, and never substitute a CPU lane. None of these implementation-level improvements qualifies a source-native PKLSA clock.